Cell-Free Massive MIMO Channel Prediction

Literature-Grounded Study with Baseline and Ablation Analysis

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Presentation Roadmap

1. Literature Survey and Motivation
2. Initial Understanding
3. Baselines
4. Our Approaches and Experiments
5. Comparison with Baselines
6. Understanding of Results
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Literature Survey: Key Takeaways

- Massive MIMO and Cell-Free MIMO improve coverage, but CSI aging degrades beamforming under mobility.
- Classical predictors (AR, Kalman/VKF) are interpretable and strong baselines, but can be computationally heavy.
- Recent DL methods (CNN, LSTM, Transformer) improve prediction accuracy in dynamic channels.
- Literature survey highlights a gap: lightweight predictors for **distributed, irregular AP topologies** in Cell-Free systems.
- Our project targets this gap with a temporal CNN and a GNN-CNN hybrid, tested against AR and VKF baselines.

Motivation and Research Question

Motivation from literature survey and report:

- Channel estimates become stale during pilot-to-data delay, especially at high velocity.
- Centralized methods are accurate but can add latency and scale poorly.
- Edge-ready 6G systems need low-latency prediction with strong robustness.

Research question:

Can a decentralized temporal model outperform both classical predictors and centralized graph aggregation for multi-step channel prediction in Cell-Free Massive MIMO?

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Our Initial Understanding (from notebooks + early report)

- Initial hypothesis: neighboring APs should share useful correlation, so GNN aggregation should help prediction.
- Early pipeline in notebooks: dataset generation, train CNN-only and GNN-CNN, evaluate by velocity, step, ablations, and efficiency.
- System configuration used across experiments: $L = 16$ APs, $N = 4$ antennas/AP, $K = 4$ UEs, history $T_h = 10$, horizon $T_p = 3$.
- Velocity settings: 30, 60, 120 km/h; layouts: random and grid.
- Key insight that changed trajectory: normalization and physically grounded channel generation strongly impact model behavior.

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Procedure Overview: End-to-End Pipeline

1. Define simulation settings: AP/UE counts, mobility, topology, history horizon, prediction horizon.
2. Generate channel dataset with geometry-aware fading model.
3. Build graph connectivity (KNN) for the GNN-CNN path.
4. Train models (CNN-Only and GNN-CNN) for each layout and velocity setup.
5. Test against AR and VKF baselines on held-out data.
6. Run full evaluation suite and export plots/JSON summaries.

Procedure Step 1: Dataset Generation

- AP layouts: random and grid; user mobility at 30/60/120 km/h.
- Large-scale fading: pathloss + spatially correlated shadowing.
- Small-scale fading: geometry-based multi-path temporal evolution.
- Sample format: input history $X \in \mathbb{R}^{L \times 2NK \times T_h}$, target $Y \in \mathbb{R}^{L \times 2NK \times T_p}$.
- Dataset split used for experiments: train/validation/test.

Procedure Step 2-3: Training and Testing

Training stage

- Train both neural models for each (layout, velocity) configuration.
- Use per-AP normalization in the forward path for stable learning.
- Save checkpoints per model/configuration for reproducible evaluation.

Testing and analysis stage

- Evaluate neural models and classical baselines on the test split only.
- Metrics and studies: NMSE vs velocity, NMSE vs step, GCN ablation, topology ablation, efficiency.
- Export artifacts: JSON metric files and publication-ready result plots.

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Baseline 1: AR Predictor

- Scalar AR(4) per channel coefficient using Yule-Walker estimation.
- Predicts future values autoregressively from local history only.
- Strength: simple, low parameterization, easy to interpret.
- Limitation: weak under non-stationary/high-mobility channel evolution.

Baseline 2: Vector Kalman Filter (VKF)

- Joint state tracking across AP antenna dimensions with covariance updates.
- Uses linear state-space prediction/correction and matrix inversions.
- Strength: principled statistical tracker with good physical interpretability.
- Limitation: computational overhead and sensitivity to model mismatch.

Baseline Summary Table

Method	Core idea	Strength	Practical limitation
AR(4)	Per-link autoregression from recent history	Very lightweight and fast to deploy	Misses richer temporal/spatial structure
VKF	Joint vector state prediction with covariance update	Strong classical benchmark and interpretable	High latency due to matrix operations

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Approach 1: CNN-Only (Decentralized)

- Shared temporal Conv1D stack per AP, followed by prediction head.
- Per-AP normalization and de-normalization in forward pass.
- No AP-to-AP message passing: purely local temporal representation learning.
- Designed for low-latency decentralized deployment.

Approach 2: GNN-CNN (Centralized Aggregation)

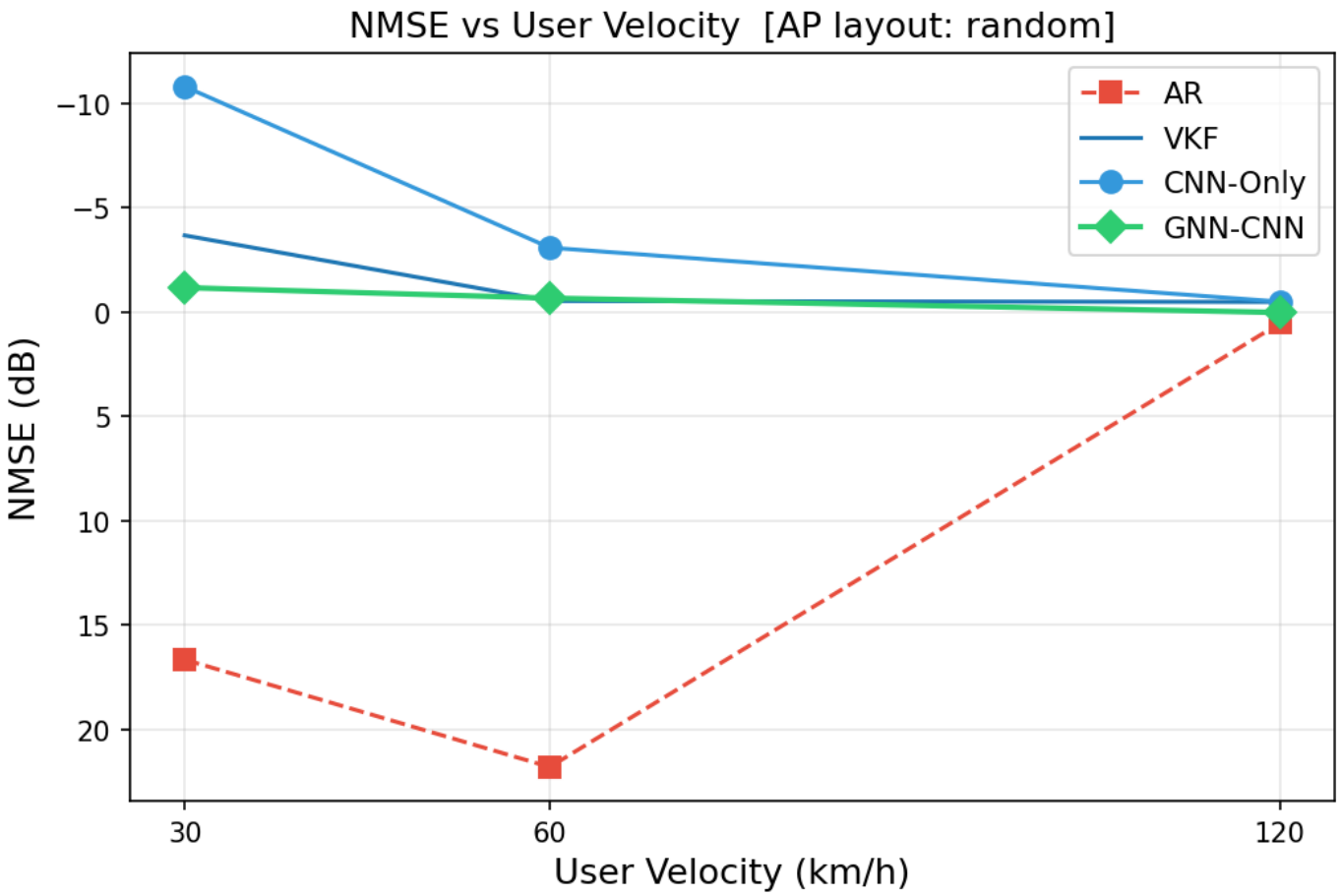
- Stage 1: local temporal CNN embedding per AP.
- Stage 2: single GCN layer over KNN AP graph ($k = 3$) to aggregate neighborhood context.
- Stage 3: prediction head for multi-step channel output.
- Goal: test whether explicit spatial aggregation improves prediction under mobility.

All Experimentations Performed

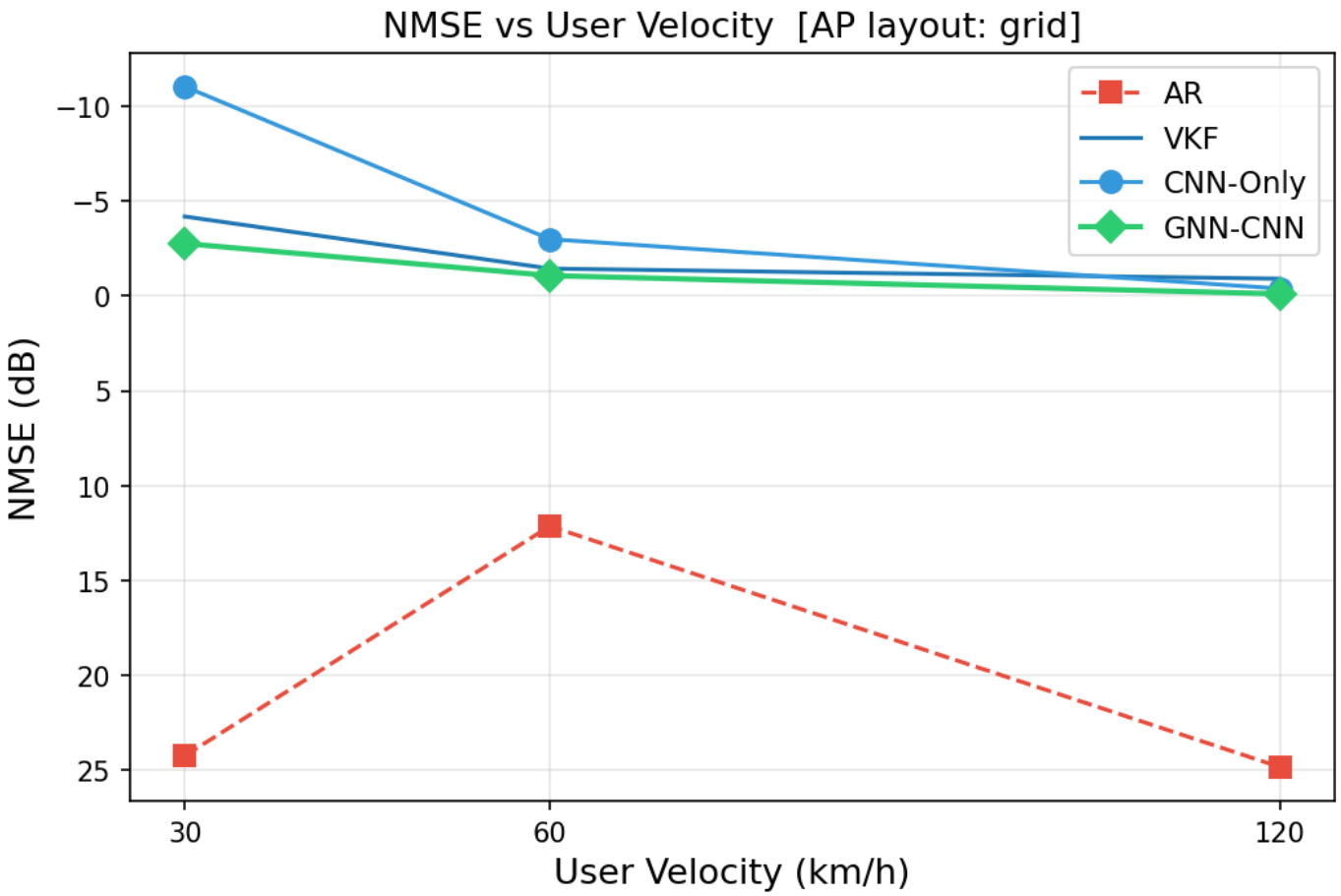
Exp.	Name	Objective
1	NMSE vs velocity	Compare AR, VKF, CNN-Only, GNN-CNN at 30/60/120 km/h
2	NMSE vs step	Evaluate error growth for $k = 1, 2, 3$ ahead prediction
3	GCN ablation	Compare CNN-Only vs GNN-CNN directly
4	Topology ablation	Compare random vs grid AP layout impact
5	Efficiency	Compare parameters and inference latency

- Iteration progression used in report: initial model $\rightarrow$ normalization fix $\rightarrow$ GBSM + VKF verification.

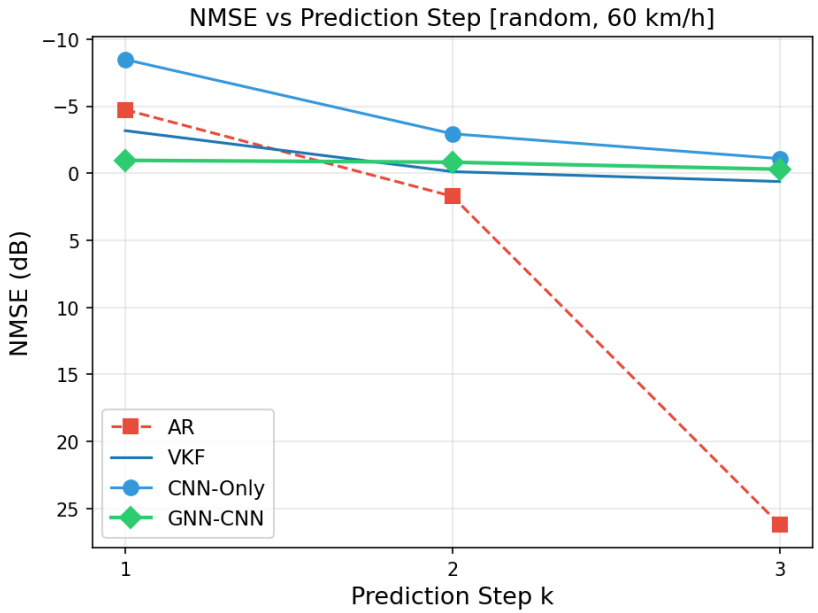
Result Plot: NMSE vs Velocity (Random Layout)



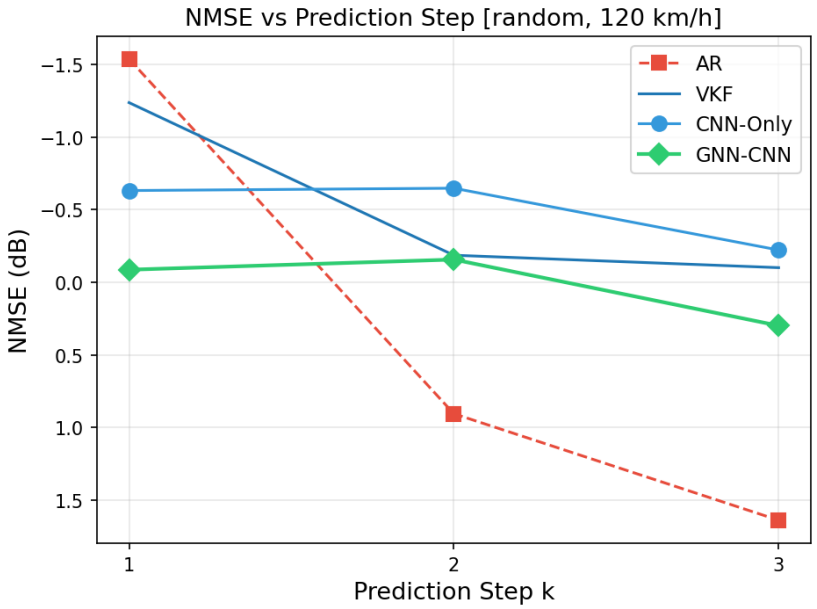
Result Plot: NMSE vs Velocity (Grid Layout)



Result Plot: NMSE vs Prediction Step

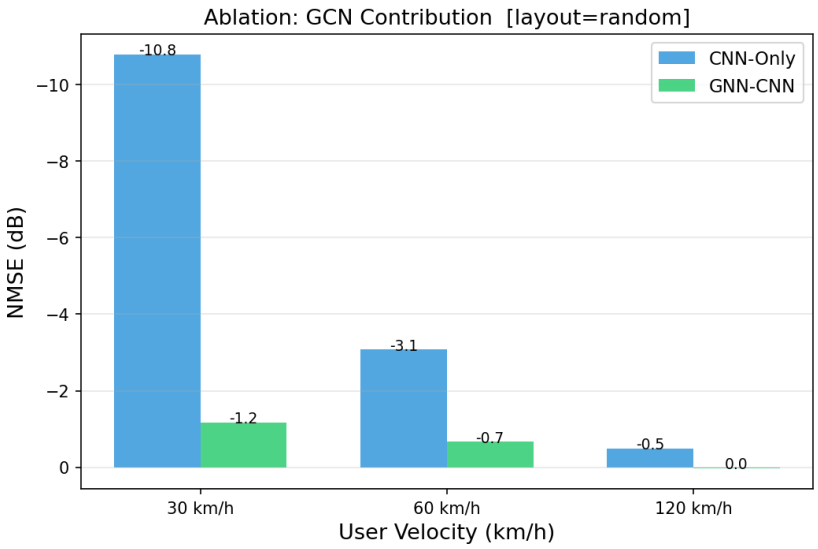


60 km/h (random)

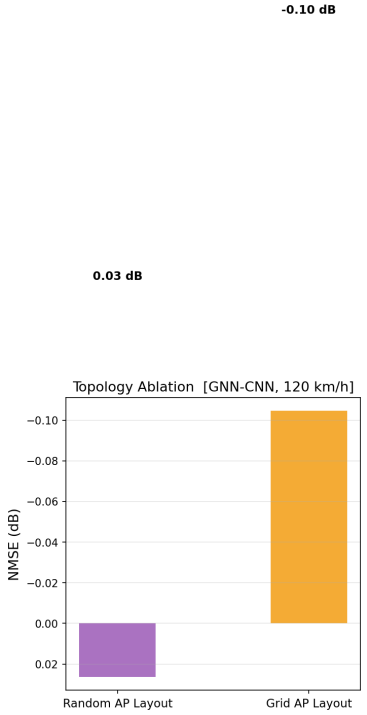


120 km/h (random)

Result Plot: GCN Ablation and Topology Ablation



CNN-Only vs GNN-CNN



Grid vs random (120 km/h)

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Comparison Table: NMSE vs Velocity (Random Layout)

Method	30 km/h	60 km/h	120 km/h
AR	16.65	21.79	0.54
VKF	-3.67	-0.52	-0.48
CNN-Only	**-10.79**	**-3.08**	**-0.49**
GNN-CNN	-1.16	-0.66	0.03

More negative NMSE is better.

Comparison Table: NMSE vs Velocity (Grid Layout)

Method	30 km/h	60 km/h	120 km/h
AR	24.26	12.15	24.84
VKF	-4.19	-1.44	-0.90
CNN-Only	**-11.04**	**-2.99**	-0.39
GNN-CNN	-2.76	-1.07	**-0.10**

Efficiency Table: Practical Deployment View

Method	Params	Mean latency (ms)	Std (ms)
GNN-CNN	51008	0.928	0.146
CNN-Only	46848	**0.712**	0.038
AR	1024	9.639	0.359
VKF	256	35.972	1.037

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Understanding the Results

- CNN-Only consistently beats VKF and often beats GNN-CNN, especially at 30/60 km/h.
- Per-AP normalization was the turning point; without it, neural models underperformed.
- GNN aggregation can mix uncorrelated AP innovations, hurting prediction in some settings.
- Grid layout slightly helps GNN-CNN at very high speed (120 km/h), indicating topology still matters.
- For edge deployment, CNN-Only gives the best overall accuracy-latency trade-off.

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Conclusion

- We started with a centralized-correlation hypothesis (GNN-CNN expected to win).
- Through iterative experimentation (modeling fixes, normalization, GBSM, VKF benchmark), we observed the opposite.
- A decentralized temporal CNN achieved the strongest and most stable performance.
- Final takeaway: for this project setting, local temporal learning is more effective and more deployable than heavier centralized aggregation.

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References I

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Thank You